
CUMMINS



FAULT CODE: 3575

PID(P):

SPN: 4334

FMI: 0/16

LAMP: AMBER

AFTERTREATMENT 1 DIESEL EXHAUST FLUID
PRESSURE – DATA VALID BUT ABOVE
NORMAL OPERATING RANGE – MODERATELY
SEVERE LEVEL

REASON

The diesel exhaust fluid (DEF) dosing unit has detected a blockage in the diesel exhaust fluid return flow.

EFFECT

Reduced engine performance is possible.

CIRCUIT DESCRIPTION

A 5-volt supply and a ground are supplied to the DEF pressure sensor by the ECM. The pressure sensor then provides a signal to the ECM on the sensor signal circuit. This signal voltage changes, based on the DEF pressure supplied by the dosing unit. A low signal voltage will indicate low DEF pressures, and a high signal voltage will indicate high DEF pressures.

COMPONENT LOCATION

The DEF pressure sensor is internal to the DEF dosing unit and must be serviced as a single component.

CONDITIONS FOR RUNNING THE DIAGNOSTICS

When the key is in the on position or the engine is running, this diagnostic runs continuously.

CONDITIONS FOR SETTING THE FAULT CODES

The DEF pressure was greater than 159 psi (1100 kPa) for more than 20 seconds.

ACTION TAKEN WHEN THE FAULT CODE IS ACTIVE

When the diagnostic runs and fails on two consecutive trips, the amber check engine light or MIL will turn on, if the engine is operated for an extended period with this fault code active, the engine torque will be severely reduced and the vehicle speed will be limited to 5 mph (8 kph).



CONDITIONS FOR CLEARING THE FAULT CODE

After the diagnostic runs and passes, the check engine light or MIL will turn off. For OBD, after 3 consecutive trips with the diagnostic running and passing, the MIL will turn off. The reset all faults command in the diagnostic tool can be used to clear active and inactive faults, as well as extinguish the MIL for OBD applications. To validate the repair, connect a diagnostic tool and perform the aftertreatment DEF doser pump override test.

SHOP TALK

Troubleshoot other active SCR related faults before checking this fault. A blocked, restricted, or frozen DEF dosing unit return line or return line fitting or a malfunctioning DEF dosing unit can be causes of this fault code. Use the appropriate OEM wiring diagrams when troubleshooting OEM wiring circuits. It is highly recommended to check the OEM power distribution center fuses, relays and connections thoroughly when troubleshooting intermittent power supply and datalink communication issues with aftertreatment components.

TROUBLESHOOTING STEPS

Step 1) Check for primary fault codes

Conditions

- Turn the key switch to the on position.

Action

- Connect a diagnostic tool and read the fault codes.

Specification

- Troubleshoot other fault codes from the following list that is or has been active within the last 25 hours before troubleshooting this fault code:
- Fault code: 3571, 3572, 3596, 3713, 4167, 5278

Repair

- Follow the steps as stated in the appropriate repair plan as necessary.

Step 2) Aftertreatment diesel exhaust fluid dosing valve is malfunctioning

Conditions

- Turn the key switch to the off position.

Action

- Connect a diagnostic tool and perform the aftertreatment DEF doser pump override test.

Specification

- If the test is not within specification, follow the instructions in the repair section.

Repair

- Replace the aftertreatment diesel exhaust fluid dosing valve as necessary.

Validation

- Follow the steps as outlined in the overview section. Check the ECM calibration and revisions for any applicable fixes or updates and recalibrate the ECM as necessary.



Step 3) Aftertreatment diesel exhaust fluid dosing unit line or fittings are cracked, plugged or restricted

Conditions

- Turn the key switch to the off position.

Action

- Check and be sure the lines are not swapped at the dosing unit, dosing valve or DEF tank connections.
- Check the DEF tank and both dosing unit connections and fittings for cracks, pitting, looseness, leaks or damage.
- Disconnect the suction and pressure line dosing unit connectors and check for any build-up or blockage.
- Check the suction line for any leakage, cracks or openings that may allow air to suck into the system.
- Check the pressure line for any damage, cuts, cracks, ruptures or signs of leakage.
- Be sure all the connection points, including the DEF tank, dosing unit, and dosing valve are sealed properly.

Specification

- Follow the instructions in the repair section if any damage, restrictions, leaks, or the lines reversed is found.

Repair

- Repair or replace the reversed, damaged, restricted or failed component as necessary.

Step 4) ECM calibration revision history check

Conditions

- Connect all components that may be disconnected.
- Turn the key switch to the on position.
- Connect a diagnostic tool.

Action

- Check the ECM calibration and revisions for any applicable fixes or updates.

Specification

- If a calibration update is available, follow the information in the repair section below.

Repair

- Document all job images and any troubleshooting information before downloading the updated ECM calibration code. The calibration download will remove the fault codes.

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Validation

- Follow the steps as outlined in the overview section. Check the ECM calibration and revisions for any applicable fixes or updates and recalibrate the ECM as necessary.

Step 5) Aftertreatment diesel exhaust fluid dosing unit is malfunctioning

Conditions

- None.

Action

- Verify all steps have been correctly performed to correct this fault.

Specification

- If the fault code is still active after all steps of troubleshooting have been performed correctly, then a faulty DEF dosing unit has been detected.

Repair

- Replace aftertreatment diesel exhaust fluid dosing unit as necessary.

Validation

- Follow the steps as outlined in the overview section. Check the ECM calibration and revisions for any applicable fixes or updates and recalibrate the ECM as necessary.
- If no root cause has been found, return to the beginning of the troubleshooting steps. After the steps have been taken again without a resolution, contact your local authorized repair facility.